Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3341-WE01

Title:		
MEng/BEng: Environmental Engineering 3		
Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

1)

a) What is the definition of sustainable development?

(5%)

b) List one physical, one biological and one chemical attribute of a healthy soil which is important to civil engineers to ensure sustainable built infrastructure

(15%)

c) In relation to contaminated land:

(i) explain what SPOSH is;

(ii) describe the two processes used to ensure that land is safe for purpose.

(40%)

d) In relation to where land is contaminated with both heavy fuel oil and petrol:

(i) describe the methods used to detect and establish the extent of contamination;

(ii) suggest methods for remediation of both types of pollution.

(40%)

2)

a) All solutions are electrically neutral. How can you use this fact, when presented with water quality data from a site investigation, to determine whether or not you trust that data.

(15%)

b) If you are building a housing estate very close to a river part of whose riverbed is coated with an orange deposit, what investigations might you undertake to establish the impact of this pollution and what threat it poses to your site, both in terms of people's health and the sustainability of the built infrastructure?

(20%)

continued

c) If the river water has a pH of 5.5 and little or no HCO_3^-, explain what treatment system you might use to clean polluted water entering the river? Explain what assumptions you have made and use diagrams as well as reference to pH and redox to illustrate your answer.

(65%)

3) A hydraulic jump is observed in a horizontal open channel of rectangular cross-section with a breadth of 300 cm. The water depths upstream and downstream of the jump are 30 cm and 130 cm, respectively.

a) Use a combination of the continuity equation and the momentum equation to determine the volumetric flow rate in the channel.

(50%)

b) Use the Bernoulli equation to subsequently determine the power dissipated within the hydraulic jump.

(50%)

END